Persona: Sam J. 32 - Medical Lab Technician (Primary User)

Goal: Register a new patient sample, process it through the workflow, enter the test result, and confirm the system automatically flags it as abnormal.

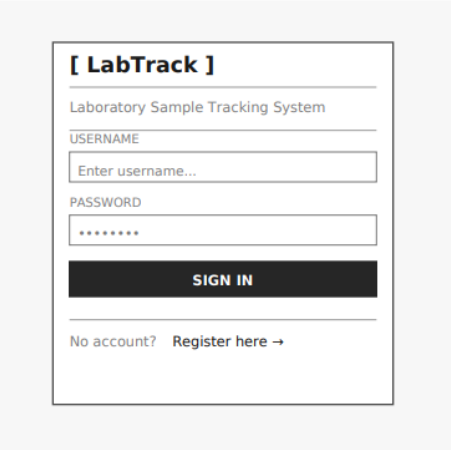
Scenario: It's Monday morning. A Glucose sample for patient John Doe has just arrived. Sam opens LabTrack to register it and process it through to completion.

Story Flow Overview:

1. Login → 2. Dashboard → 3. New Sample → 4. Status → 5. Enter Result → 6. Flag Assigned

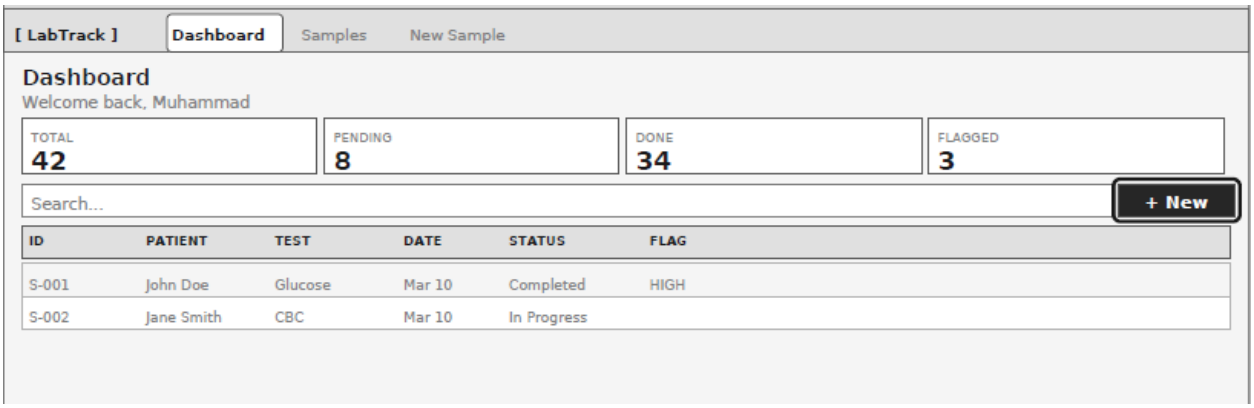
STEP 1 & 2 - Login and Dashboard

1. Action: Sam opens LabTrack in her browser and enters her username and password.



Result: Credentials verified → Spring Security creates session → redirected to /dashboard

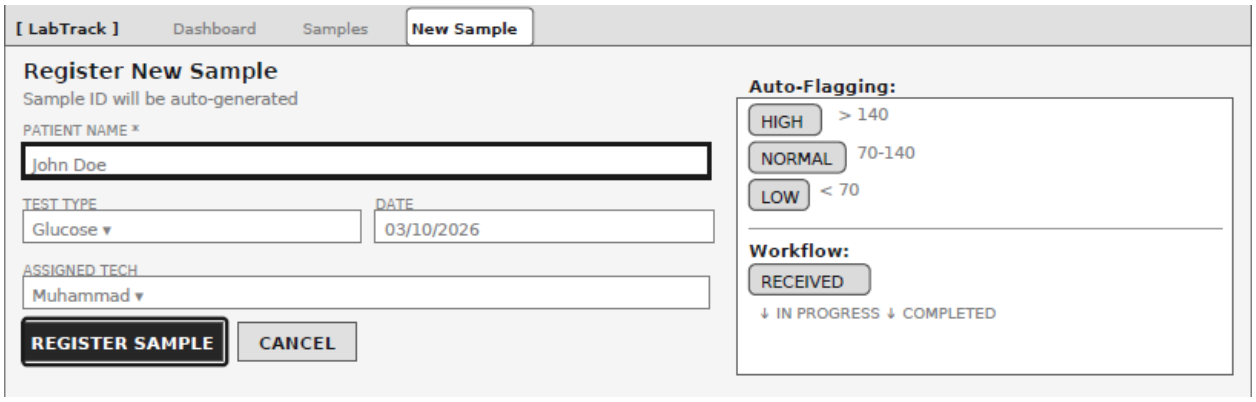
2. Action: Sam lands on the Dashboard. She sees 8 pending samples and clicks '+ New' to register a new one.



Result: Browser navigates to /samples/new - New Sample form

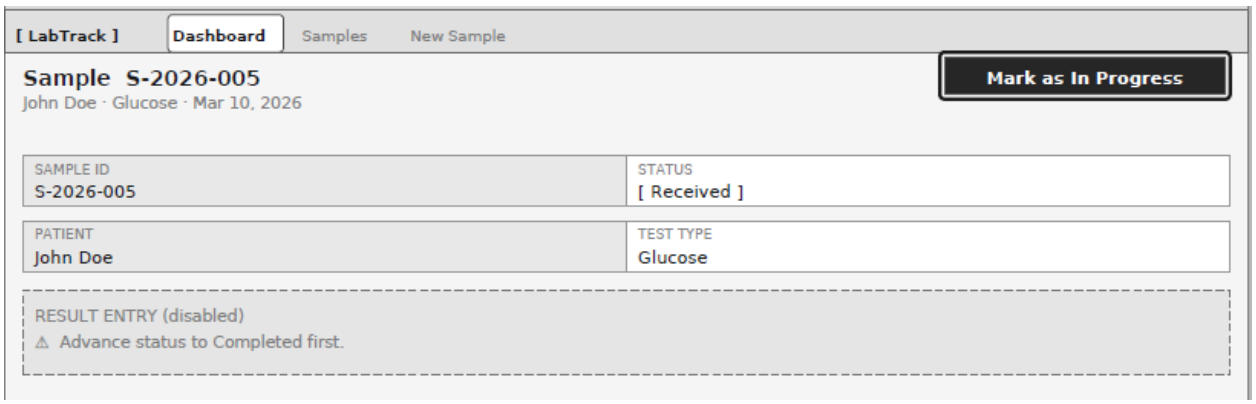
STEP 3 & 4 - Register Sample and Status

3. Action: Sam fills in the form: Patient = John Doe, Test Type = Glucose, Date = Mar 10. Clicks Register Sample.



Result: Sample S-2026-005 created with status = RECEIVED → redirected to Sample Detail page

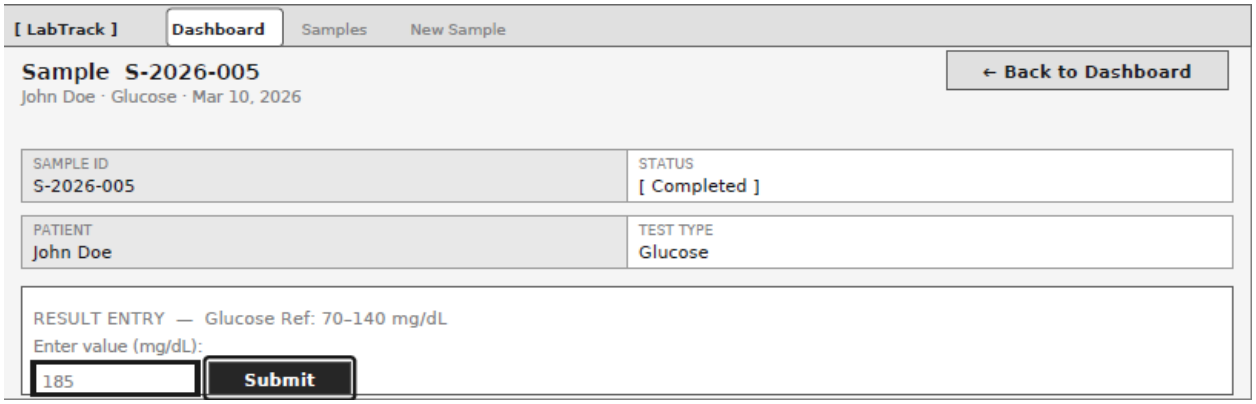
4. Action: Sam sees the Sample Detail page. She clicks 'Mark as In Progress', then 'Mark as Completed'.



Result: Service layer validates transition → status updated to IN PROGRESS then COMPLETED → page reloads

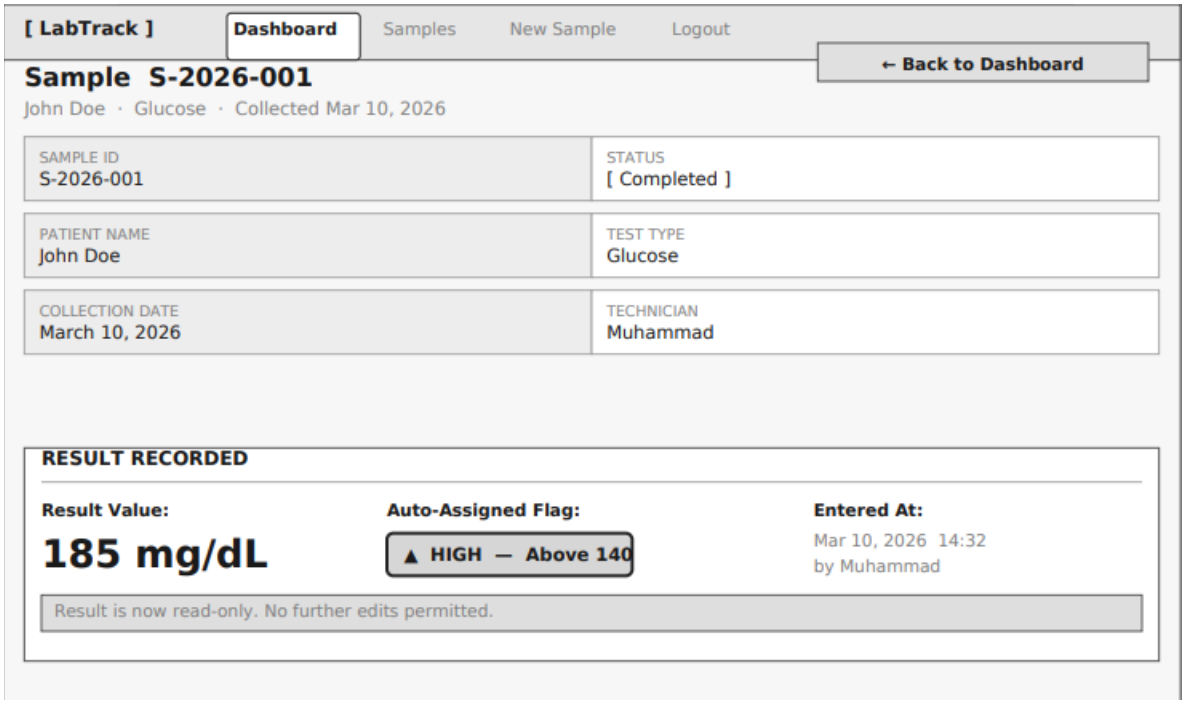
STEP 5 & 6 - Enter Result and View Auto-Assigned Flag

5. Action: Sample is now Completed. Sam enters the result value 185 in the result field and clicks Submit.



Result: Service layer receives 185 → compares against Glucose max (140) → 185 > 140 → assigns flag = HIGH

6. Action: Page reloads. Sam immediately sees the result recorded with a HIGH flag. No manual tagging needed.



Result: Goal Completed – Sam registered, processed, and got an auto-flagged result without any manual judgement.